

Equal-Count Injectivity of Noncontracting Prefixes Through 31 Steps

Collatz proof project — September 5, 2026

Abstract

For every time at most 31 and every pair of natural seeds, noncontracting prefixes with the same odd count and endpoint have the same seed. Lean proves this using a small correction-envelope certificate, independently of the exploratory enumeration. We also certify why the odd-count hypothesis is essential and why this particular correction bound stops at 31. No arbitrary-time injectivity or Collatz theorem is claimed.

Use the shortcut map $T(n) = n/2$ for even n , and $(3n+1)/2$ for odd n . Write $j_t(n)$ for the odd count and $d_t(n)$ for the affine correction:

$$2^t T^t(n) = 3^{j_t(n)} n + d_t(n).$$

A prefix through t is noncontracting if $2^i \leq 3^{j_i(n)}$ for every $0 \leq i \leq t$.

Theorem 1 (Correction interval). *If $t \leq 31$ and the prefix from n through t is noncontracting, then*

$$3^{j_t(n)} \leq d_t(n) + 2^{j_t(n)} < 5 \cdot 3^{j_t(n)}.$$

There is no bound on n .

Proof. The lower inequality follows from the repository's general lower master inequality and the exact affine form. For the upper inequality, let $H_{0,j} = 0$. For $0 \leq j \leq t+1$, set

$$H_{t+1,j} = \begin{cases} \max\{H_{t,j}, 3H_{t,j-1} + 2^t\}, & 2^{t+1} \leq 3^j, \\ 0, & \text{otherwise,} \end{cases}$$

with the second candidate omitted when $j = 0$ and out-of-range entries zero. The two candidates bound an appended even or odd step respectively. Induction along an actual noncontracting prefix gives $d_t(n) \leq H_{t,j_t(n)}$. A 32×32 natural-number table is supplied to Lean. The kernel checks its base value, both transition inequalities, and $H_{t,j} + 2^j < 5 \cdot 3^j$ for $j \leq t \leq 31$ with $2^t \leq 3^j$. These checked inequalities prove the claimed upper bound. $\square$

Theorem 2 (Equal-count injectivity). *Suppose $t \leq 31$, both seeds n, m have noncontracting prefixes through t , $j_t(n) = j_t(m)$, and $T^t(n) = T^t(m)$. Then $n = m$.*

Proof. For $t \geq 2$, the first two steps of each prefix must be odd: an even first step or an odd-even start contracts the coefficient. Thus $n \equiv m \equiv 3 \pmod{4}$. Write j for their common odd count. The affine identities imply

$$3^j(n - m) = d_t(m) - d_t(n).$$

The correction interval gives $|n - m| < 4$. Congruence modulo four then forces equality. Time zero is immediate; at time one the map is injective on the forced odd seeds. $\square$

Two distinct limits of the statement. The odd-count hypothesis cannot be dropped: Lean checks $T^7(31) = T^7(95) = 182$, with odd counts six and five respectively, and both prefixes noncontracting. The displayed correction interval cannot simply be extended to time 32 either. For $n = 3384695803$, Lean checks all noncontraction inequalities through 32 and

$$j_{32}(n) = 21, \quad d_{32}(n) = 54020229503, \quad 5 \cdot 3^{21} \leq d_{32}(n) + 2^{21}.$$

This is a counterexample to the strict interval, not to injectivity at 32.

Research consequence and verification. There is no distinct equal-time/equal-count replacement through 31 steps when both prefixes must remain noncontracting. This explains a limitation of the dominating-merge route; it does not exclude unequal counts or unequal lengths. The module is `Collatz.NCPrefixInjective`, with four public declarations in the axiom audit. The generator is `analysis/nc_correction_envelope.py`. An independent Python search found no equal-count collision through depth 30 (12,771,274 prefixes at that depth); it is exploratory evidence, not an input to the proof. An extension needs stronger relative correction information or another argument. Collatz remains open. No mathematical priority claim is made.